Perfectization and perfectoid crystals.

Setup

In this talk:

1) $\mathfrak{X}$ is a bounded$^{(*)}$ p-adic formal scheme. This means that $\mathfrak{X} = \bigcup \mathrm{Spf}(R)$ in the Zar topology where R is a p-complete and bounded p^∞-torsion

"p-adic formal scheme" will mean "bounded" p-adic formal.

(*) This means that $\exists n \in \mathbb{N}$ s.t if $p^m r = 0$ in R for some $m \gg 0$ then $p^n r = 0$. (Automatic if R Noetherian)

The main benefit from boundedness is that derived p-completion

$\|$

classical p-completion.

2) SPfd^{nil} is the category of p-nilpotent semiperfectoid rings.

It is a basis of p-nilpotent rings for the flat topology.

↗ Every object will be seen through the lens of it's functor of points on SPfd^{nil}

e.g a) S p-nilpotent:

$$\mathrm{Spec}(S) : \mathrm{SPfd}^{nil} \longrightarrow \mathrm{Set}$$

$$A \longmapsto \mathrm{Ring}(S, A)$$

b) R p-complete

$$\mathrm{Spf}(S) := \varinjlim_n \mathrm{Spec}\left(R/p^nR\right)$$

(in (pre)sheaves)

1) Untilts and perfectization

Def $S \in \mathrm{SPfd}^{nil}$. An untilt of S is

$$S^\# \longrightarrow S$$

$S^\#$ perfectoid, with $(S^\#)^\flat \xrightarrow{\sim} S^\flat$.

□

Remark: Automatically $S^\# \twoheadrightarrow S$. (exercise)

Remark: Equivalently, choice of (perfect) prism structure $(W(S^\flat), I)$ s.t.
$I \subseteq \ker(\theta_S : W(S^\flat) \longrightarrow S)$ (exercise)

Def For any presheaf (of anima) $\mathcal{X}$ on SPfd^{nil} define the presheaf

$$\mathcal{X}^{pfd}(S) = \{ \underset{\text{untilt}}{\mathrm{Spec}(S) \hat{\to}} \mathrm{Spf}(S^\#) \underset{\text{a map}}{\hat{\to}} \mathcal{X} \}$$

Properties. 0) There is a natural map $\pi : \mathcal{X}^{pfd} \longrightarrow \mathcal{X}$. □

1) Discreteness

Automorphisms of untilts are $\{id\}$

If

$$\begin{tikzcd} S^\# \arrow[d, "\sigma"', "\wr"] \arrow[dr, "f"] & \\ S^\# \arrow[r, "f"] & S \end{tikzcd}$$

automorphism of untilts, then:

$$\begin{tikzcd} (S^\#)^\flat \arrow[d, "\sigma^\flat"'] \arrow[dr, "f^\flat", "\sim"'] & \\ (S^\#)^\flat \arrow[r, "f^\flat", "\sim"'] & S^\flat \end{tikzcd}$$

$\sigma^\flat = id \Longleftarrow$

$\Downarrow$ untilting

$\sigma = id$.

So if $\mathcal{X}$ presheaves of sets, $\mathcal{X}^{pfd}$ also.

2) $(-)^{pfd}$ preserves fiber products and direct limits (exercise)

3) If $\mathcal{X}$ flat sheaf, so is $\mathcal{X}^{pfd}$

4) If $\mathcal{U} \to \mathcal{X}$ formally étale

$$\mathcal{U}^{pfd} \xrightarrow{\sim} \mathcal{X}_{pfd} \times_{\mathcal{X}} \mathcal{U}$$

By def theory for $S \in \mathrm{SPfd}^{nil}$

$$\begin{tikzcd} \mathrm{Spec}(S) \arrow[r] \arrow[d] & \mathcal{U} \arrow[d] \\ \mathrm{Spf}(S^\#) \arrow[r] \arrow[ur, dashed, "\text{unique}"] & \mathcal{X} \end{tikzcd}$$

5) If R is pfd then

$$\mathrm{Spf}(R)^{pfd} \xrightarrow{\sim} \mathrm{Spf}(R)$$

Proof Let $S \in \mathrm{SPfd}^{nil}$.

$$\mathrm{Spec}(S) \longrightarrow \mathrm{Spf}(R)^{pfd}$$

is

$$\begin{tikzcd} R \arrow[r] \arrow[dr] & S^\# \arrow[d, "\text{untilt}"] \arrow[dl, phantom, "\circlearrowleft"] \\ & S \end{tikzcd}$$

if this commutes then

$$(A_{inf}(R), \ker\theta) \longrightarrow (W(S^\flat), I)$$

with $I = \ker \theta W(S^\flat)$

necessarily by rigidity of prims.

So there is no choice of untilt.

Also the map $R \longrightarrow S^\#$ has to tilt to the map $R^\flat \longrightarrow S^\flat$ so it is also uniquely determined. □

6) If R is pfd then

$$\mathcal{X}^{pfd}(R) = \mathcal{X}(R)$$

Proof Have to show by def

$$\varprojlim_n \mathcal{X}^{pfd}(R/p^n) = \varprojlim_n \mathcal{X}(R/p^n)$$

But the (groupoid of) compatible collections of untilts $\{ S \xrightarrow[\text{untilt}]{} R/p^n \}_n$ consits (up to unique isomorphism) of $\{ R \to R/p^n \}_n$ □

7) By 6) and inspection of the definition,

$\mathcal{X} \longrightarrow \mathcal{Y}$ morphism of p-adic formal

- induces $\mathcal{X}^{pfd} \xrightarrow{\sim} \mathcal{Y}^{pfd}$

$\Longleftrightarrow$

- $\forall R$ pfd $\mathcal{X}(R) \xrightarrow{\sim} \mathcal{Y}(R)$.

8) Say S is a ring s.t. there exists $S \longrightarrow S_{pfd}$ an intial map to a perfectoid ring.

Then

$$\begin{tikzcd} \mathrm{Spf}(S_{pfd}) \arrow[r] \arrow[d, leftrightarrow, "\wr"] & \mathrm{Spf}(S) \\ \mathrm{Spf}(S)^{pfd} \arrow[ur] & \end{tikzcd}$$

by 7).

Proposition Let $\mathcal{X} \longrightarrow \mathcal{Y}$ be finite (closed). Then $\mathcal{X}^{pfd} \to \mathcal{Y}^{pfd}$ is affine (closed).

Proof Let $S \in \mathrm{SPfd}^{nil}$. Recall

$$\begin{tikzcd} \mathrm{Spec}(S) \arrow[r, "\text{untilt}"] & \mathrm{Spf}(S^\#) \arrow[r] & \mathcal{Y} \\ & \arrow[u, leftrightarrow, "\text{def}"] & \\ \mathrm{Spec}(S) \arrow[rr] & & \mathcal{Y}^{pfd} \end{tikzcd}$$

By alm purity (Bhatt's seri theorem)

Consider:

$$\begin{tikzcd} \mathcal{Z} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d] \arrow[r, squiggly, "(-)^{pfd}"] & \mathcal{Z}^{pfd} \arrow[r] \arrow[d, "(*)"] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X}^{pfd} \arrow[d] \\ \mathrm{Spec}(S^\#) \arrow[r] & \mathcal{Y} & \mathrm{Spec}(S^\#) \arrow[r] & \mathcal{Y}^{pfd} \end{tikzcd}$$

(*) is affine (closed). □

Examples

1) $\mathbb{Z}_p^{pfd}(S) = \{ \text{untilts of } S \}$

2) $\underset{\sim}{\mathbb{G}_{m,R}^{pfd}}$ Fix R p-tors free pfd ring

Have

$$\begin{tikzcd} \mathbb{G}_{m,R}^{Tate} := \mathrm{Spf}(R\langle x^{\pm 1/p^\infty}\rangle) \arrow[d] \\ \mathbb{G}_{m,R} = \mathrm{Spf}(R\langle x^{\pm 1}\rangle) \end{tikzcd}$$

flat surjection of p-adic formal group schemes.

Kernel: $\mathbb{Z}_p(1)_R$; on points get

$$1 \to \mathbb{Z}_p(1)_R(S) \to S^{\flat\times} \longrightarrow S^\times \to 1$$

$$\mathbb{Z}_p(1)_R(S) = \{ (\zeta_{p^n}) \in S^{\flat\times} \mid \zeta_{p^0} = 1 \}$$

$$\underset{\substack{\uparrow \\ \text{semiperfectoid.}}}{\mathbb{Z}_p(1)_R} = \mathrm{Spf}\left(R\langle x^{\pm 1/p^\infty}\rangle / (x-1) \right)$$

$$\mathbb{Z}_p(1)_R^{pfd} = \mathrm{Spf}\left(R\langle x^{\pm 1/p^\infty}\rangle / (x-1)_{pfd} \right)$$

a "smaller subgroup scheme" of $\mathbb{G}_{m,R}^{Tate}$.

One can show

$$\mathbb{G}_m^{pfd} = \mathbb{G}_{m,R}^{Tate} / \mathbb{Z}_p(1)_R^{pfd}.$$

Why the word "un"tilt?

(In a way) the "tilt" of a perfectoid ring R is entirely captured by R/p (as $R^\flat = \varprojlim_{(-)^p} R/p$)

Now let $S = R/p$. Then, there is as many "untilts"

$$S^\# \longrightarrow S$$

as perfectoid rings S and maps

$S^\# \longrightarrow R/p$ that induces an isomorphism modulo p.

for $S = R/p$,

$$\mathbb{Z}_p^{pfd}(S) = \left\{ \begin{tikzcd} S^\# \arrow[d, "\text{untilts}"] \\ S \end{tikzcd} \right\}$$

perfectoids rings with a map to S which is an iso mod p

2) Perfectoid atlases

Theorem Let $\mathcal{X}$ p-adic formal scheme. Then $\exists$

$$\begin{tikzcd} \mathcal{U} \arrow[r] & \mathcal{X} \end{tikzcd}$$

pfd formal scheme

flat cover.

Proof $\mathcal{X} = \mathrm{spf}(R)$ wlg.

Say $\mathcal{X} \hookrightarrow \mathcal{Y}$ s.t thm is showed for $\mathcal{Y}$. Then

$$\begin{tikzcd} \mathcal{U} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{U}' \arrow[d] \\ \mathcal{X}^{pfd} \arrow[r, hook] & \mathcal{Y}^{pfd} \end{tikzcd}$$

gives the general assertion by base change.

So the case of $\mathbb{Z}_p\langle x_i\rangle_{i\in I}$ suffices. Let

$$\begin{tikzcd} \mathcal{U} = \mathrm{spf}\left(\mathbb{Z}_p\langle p^{1/p^\infty}, x_i^{1/p^\infty}\rangle_{i\in I}\right) \arrow[d] \\ \mathcal{X} \end{tikzcd}$$

$$= \mathbb{Z}_p\langle x_i^{1/p^\infty}\rangle[t^{1/p^\infty}]^{\wedge}_{p} \big/ (t-p)$$

$\mathcal{U}$ is affine perfectoid formal.

Take $\mathcal{U} \to \mathcal{X}^{pfd}$ induced.

We claim that it is a flat cover; take $S \in \mathrm{SPfd}^{nil}$, $\mathrm{spec}(S) \to \mathcal{X}^{pfd}$.

This is $\mathrm{spf}(S^\#) \to \mathcal{X}$ $(*)$.

untilt of S

$(*)$ corresponds to $s_i \in S^\#$ $i \in I$.

Consider the pullback

$$\begin{tikzcd} \mathrm{Spf}(A) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{U} \arrow[d] \\ \mathrm{Spf}(S^\#) \arrow[r] & \mathcal{X} \end{tikzcd} \quad (**)$$

, with:

$$A = S^\#\langle t^{1/p^\infty}, x_i^{1/p^\infty}\rangle_{i\in I} \big/ (t-p, x_i - s_i)$$

The proof of André's lemma exactly says that for such pfdizations

$$S^\# \longrightarrow (A)_{pfd}$$

is p-comp. faith flat.

Taking "pfd" of $(**)$ concludes $\square$

Corollary $\mathcal{X}$ p-adic formal scheme

Take $\mathcal{U} \to \mathcal{X}$ as above. Then

$$\mathrm{Coequ}\left(\mathcal{U} \times_{\mathcal{X}^{pfd}} \mathcal{U} \rightrightarrows \mathcal{U}\right) = \mathcal{X}^{pfd}$$

pfd formal scheme

where the coequalizer is taken in flat sheaves.

So $\mathcal{X}^{pfd}$ is a "flat algebraic space in pfd formal schemes".

3) Perfectoid Crystals

Def Perfectoid crystals:

$$\mathrm{Cris}(\overline{/A}_{\mathfrak{X}}) := \mathcal{D}(\mathfrak{X}^{\mathrm{pfd}})$$

where $\mathfrak{X}$ is p-adic formal. ☐

Have $\mathcal{D}(\mathfrak{X}^{\mathrm{pfd}}) = \varprojlim_{\substack{\mathrm{Spec}(S) \to \mathfrak{X} \\ S \in \mathrm{Spfd}^{\mathrm{nil}}}} \mathcal{D}(S)$ by def.

Actually:

Prop $\mathfrak{X}$ p-adic formal.

$$\mathrm{Cris}(\overline{/A}_{\mathfrak{X}}) := \varprojlim_{\substack{\mathrm{Spf}(R) \to \mathfrak{X} \\ R \text{ pfd}}} \widehat{\mathcal{D}}(R)$$

Proof Affine case suffices. $\mathfrak{X} = \mathrm{Spf} R$

If $S \in \mathrm{Spfd}^{\mathrm{nil}}$

$\mathrm{Spec}(S) \longrightarrow \mathfrak{X}^{\mathrm{pfd}}$ point

Corresponding to $\mathrm{Spf}(S^\#) \longrightarrow \mathfrak{X}$.

$$\begin{tikzcd} R \arrow[r] \arrow[dr] & S^\# \arrow[r] \arrow[d] & S^\#/p^n \arrow[dl, dashed, "\exists"] \\ & S & \end{tikzcd}$$

$p^n \to 0$ for some n

The existence of such a facturization shows that

$$\{ R \to S^\#/p^n \}_{n, S^\# \text{ pfd}}$$

is final in the system of all $\mathrm{Spf}(R)^{\mathrm{pfd}}$ — points ☐

Corollary $\mathfrak{X} = \mathrm{Spf}(R)$. Then

$$R\Gamma(\mathfrak{X}^{\mathrm{pfd}}, \mathcal{O}) = \varprojlim_{\substack{(S \leftarrow R) \\ \text{pfd}}} S$$

in derived rings

$$= R\Gamma_{\mathrm{arc}}(\mathrm{Spf}(R), \mathcal{O})$$

as defined previously.

Proof Take Endomorphism of the unit in the above prop.

Upshort: A perfectoid crystal on $\mathfrak{X}$ is a collection

$$(\mathcal{M}_{R,f})_{R \text{ pfd},\ f: \mathrm{Spf} R \to \mathfrak{X}}$$

with $\forall$

$$\begin{tikzcd} \mathrm{Spf} R' \arrow[d, "g"'] \arrow[dr, "f'"] & \\ \mathrm{Spf} R \arrow[r, "f"'] & \mathfrak{X} \end{tikzcd}$$

$\curvearrowleft$, have

$$g^* \mathcal{M}_{R,f} \simeq \mathcal{M}_{R',f'}$$

such that identifications satisfy "higher coherences" analogous to cocycle conditions.

4) Affineness

Given R p-complete, we have defined

$$R_{pfd}$$

which is in general a non-discrete coconnective derived ring.

Using "Higher algebra" there is a notion of (p-complete) modules over R_{pfd}

$$\widehat{\mathcal{D}}(R_{pfd})$$

(up to p-completion)

Theorem Say R is topolo$\hat{g}$ically of finite over some perfectoid ring.

Then if $\mathfrak{X} = \mathrm{Spf}(R)$

$$\mathcal{D}(\mathfrak{X}^{pfd}) \simeq \mathcal{D}(R_{pfd})$$

Sketch of **Proof** Under this hypothesis there exists

$$\begin{tikzcd} & R \arrow[dl] \arrow[dr] & & \\ R_{pfd} \arrow[rr] & & S & \arrow[l, "\text{pfd ring}"'] \end{tikzcd}$$

which induces a perfectoid atlas and $R_{pfd} \to S$ is "descendable".

Descendable is a property of maps of derived rings ($\mathbb{E}_\infty$-rings) that ensures

$$\mathcal{D}(R_{pfd}) = \varprojlim_{\Delta} \mathcal{D}(S^{\hat{\otimes}^\bullet_{R_{pfd}}})$$

Where $S^{\hat{\otimes}^\bullet_{R_{pfd}}}$ is the cosimplicial object

$$S \rightrightarrows S \underset{R_{pfd}}{\hat{\otimes}} S \overset{\rightarrow}{\rightrightarrows} S \underset{R_{pfd}}{\hat{\otimes}} S \underset{R_{pfd}}{\hat{\otimes}} S (\cdots).$$

Crucially as $(-)_{pfd}$ commutes with fiber products and that $S^{\hat{\otimes}^n_R}$ is semiperfectoid we have

$$S^{\hat{\otimes}^n_{R_{pfd}}} \cong (S^{\hat{\otimes}^n_R})_{pfd}$$
$$\cong \underset{\substack{\uparrow \\ \text{global sections}}}{\mathcal{O}}((\mathrm{Spf}(S) \times^n_{\mathrm{Spf}(R)})^{pfd})$$

Now

$$\mathcal{D}(\mathrm{Spf}(R)^{pfd}) \overset{(A)}{=} \varprojlim_{\Delta} \mathcal{D}((\mathrm{Spf}(S) \times^n_{\mathrm{Spf}(R)})^{pfd})$$
$$= \varprojlim_{\Delta} \mathcal{D}((S^{\hat{\otimes}^n_R})_{pfd})$$

because $\mathrm{Spf}(S) \to \mathrm{Spf}(R)^{pfd}$ is a pfd atlas, so (A) is flat descent for $\mathcal{D}(-)$. □

Using the same "Čech-Nerve identifications", we have:

Theorem Under the same hypothesis

$$\begin{tikzcd} \mathrm{Map}_R(\mathrm{Spf}(S), \mathfrak{X}^{pfd}) \arrow[d, "\wr"] \\ \mathrm{Der}_R(R_{pfd}, S) \end{tikzcd}$$

$\forall S$ p-complete.

Calculated in ∞-cat of R-derived algebras.

In particular this Hom space is in fact discrete!